MANet: An Efficient Multidimensional Attention-Aggregated Network for Remote Sensing Image Change Detection

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***Abstract*— Deep learning has significantly advanced the change detection (CD) in remote sensing (RS) image with its excellent performance. For CD tasks, there are two critical issues. First, with scale variance of different objects in RS images, effectively aggregating multiscale features helps to generate fine-grained change objects. Second, it is critical but challenging to fully exploit the variance information between bi-temporal images to avoid pseudo-variation and region blurring. To alleviate the above issues, this article proposes an efficient multidimensional attention-aggregation network (MANet), which keeps better feature aggregation while maintaining excellent differential attention ability. This article carries three main contributions. First, we propose a multiscale asymmetric convolutional attention (MACA) module. Due to the asymmetric convolution's ability to focus on feature contours effectively, the MACA can not only aggregate multiscale features effectively, but also refine the edge information of features. Second, we propose a dual-dimensional attention (DDA) module for adaptively fusing shallow and deep features, which is used to generate rich feature representations. Third, the difference guidance (DG) module is exploited for enhancing the attention of changed regions to mitigate the influence of uncorrelated changes on the CD result. Experiments on four popular CD datasets show that our network can accomplish higher detection accuracy than the state-of-the-art networks.**

***Index Terms*— Change detection (CD), difference guidance (DG) module, dual-dimensional attention (DDA) module, multiscale asymmetric convolutional attention (MACA) module, remote sensing (RS).**

I. Introduction

CHANGE detection (CD) is an integral branch in remote sensing (RS), which aims to define changed areas between bi-temporal images of the same geographical region at various times [1]. It has been widely used in different RS fields such as land planning [2], disaster assessment [3], and resource detection [4]. With the evolution of optical sensor devices, a large amount of high-resolution data is used for CD tasks. High-resolution particularly very-high-resolution images contain more image features and ground information, but make the CD tasks more challenging.

Traditional CD methods mainly include pixel-based and object-based ones [5], [6]. The pixel-based methods first gain the change map by comparing pixels between bi-temporal images, followed by thresholding or clustering analysis to obtain the result maps. Pixel-based approaches can be specifically classified as algebraic-based approaches, classification-based approaches, and transformation-based approaches. Algebraic-based methods mainly use arithmetic operations to obtain the result maps, such as image differencing [7] and image rationing [8]. Classification-based methods obtain the change class of a bi-temporal image directly by using a trained classifier, such as support vector machine (SVM) [9], K-nearest neighbor (K-NN) [10], and decision trees [11]. Transformation-based methods are used to derive change information in bi-temporal images by transforming features into a specific feature space, like change vector analysis (CVA) [12] and the tasseled cap transformation [13]. For high-resolution and very-high-resolution graphics, pixel-based methods are likely to lead to large detection errors due to pseudo-variations such as alignment errors and illumination changes [14], [15]. To alleviate these problems, researchers have developed object-based methods for CD tasks. The object-based approaches identify groups of pixels into objects for judgment [16]. However, they still suffer from high computational cost, low detection accuracy, and sensitivity to noise because manual features cannot sufficiently represent useful features and the design of them highly relies on the classifiers.

Over recent years, the continuous upgrading of hardware and expansion of datasets have provided a rapid development opportunity for deep learning [17]. Due to the powerful feature representation capability of networks, it is extensively used for many computer vision tasks, like image segmentation and object detection [18], [19]. Many deep learning-based CD methods have been presented and have shown better performance than traditional CD methods. In addition to convolutional neural network (CNN), there are other methods applied to RS tasks. For example, generative adversarial networks (GANs) are employed for adversarial learning to

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enhance the robustness of feature maps [20]. Graph convolutional networks (GCNs) are used to model long-range spatial relationships [21]. In the field of RS, CNN-based models have greatly pushed forward CD tasks, which are commonly grouped into early fusion (EF) and late fusion (LF) approaches by varying processing of images. EF approaches use the concatenations or variance of the image as input. For example, Alcantarilla et al. [22] stitched the bi-temporal images and later used fully convolutional network (FCN) to obtain the change map. EF methods generally employ single-branch networks, such as FCN and UNet [23], [24]. Therefore, it is easier to cause the loss of bi-temporal image information in the feature extraction process. By contrast, LF methods treat images from different times as separate inputs. Siamese-based network is employed to detect the discriminative features of the image, and the change map is obtained by feature fusion or makes difference. For example, Daudt et al. [25] presented two LF methods based on the FCN model, both models extract multilevel features at the encoder and output changed regions at the decoder. In addition, Liu et al. [26] proposed a super-resolution-based CD network (SRCDNet), which combines Siamese-based network with an attention mechanism used for feature enhancement. Wang et al. [27] introduced a metric-based model to get the changed map by evaluating the distance between various feature maps. Experimental results indicate that the LF strategies can obtain better performance over the EF strategies. Due to this reason, Siamese-based network has been broadly employed and has become the benchmark network for CD tasks [28].

Although deep learning-based approaches have made enormous progress, they still have some problems. First, in some large datasets containing complex variations, such as seasonal variations, environmental variations, and illumination variations, current CD methods cannot aggregate multiscale contextual information effectively, which leads to a negative impact on detection results [29]. Thus, an approach that can efficiently extract features at different scales is in need, for extracting richer semantic information from bi-temporal images. Second, in the popular Siamese-based networks, the encoder generates different levels of features by sequential downsampling. In general, deep features contribute to locating the change object, while shallow features contain fine-grained details and change boundaries. Therefore, there is a complementary relationship between deep and shallow features, and the current methods tend to use ordinary convolution to fuse deep and shallow features, which cannot discover the intrinsic relationship properly [30]. Therefore, more effective feature fusion methods should be explored for getting more representative fused features. Third, the existing methods have not fully exploited the difference information of images in the feature extraction process, which leads to ambiguous and erroneous predictions of changed regions by the network [31]. Consequently, for better identifying the intrinsic features of the real changed regions, more approaches still need to be explored.

To alleviate the above problems, an efficient multidimensional attention-aggregation network (MANet) is presented in this article. The approach is built on Siamese architecture and consists of three main modules: the multiscale asymmetric convolutional attention (MACA) module, the dual-dimensional attention (DDA) module, and the difference guidance (DG) module.

In summary, the main contributions in this article are summarized as follows.

1) We design a novel MANet for pixel-level CD forecasting in an end-to-end way. The network uses Siamese-based structure to capture multiscale image features.
2) To augment the feature extraction capabilities of the network, we design the MACA module. The MACA module takes multiscale feature maps from bi-temporal images and applies the features as weights to multiply with the input features. Thus, a more fine-grained representation is obtained.
3) To improve the internal compactness and integrity of the fused features, we propose the DDA module in the feature fusion stage. DDA module enables an attention-weighting mechanism in both the feature map and convolutional kernel dimensions, which enables both channel and spatial adaptive attention to fuse information from both deep and shallow features, improving the efficiency of feature fusion.
4) To better discriminate the change object and background, we propose the DG module. In practice, difference image usually contains some pseudo-variable regions due to noise and illumination disturbances, so this causes some difficulties for feature learning. Therefore, we introduced the DG module that carries attention mechanism, which can significantly distinguish changed and background regions.
5) The presented MANet attains state-of-the-art performance on four widely used CDD, LEVIR-CD, SYSU-CD, and BCDD datasets, with the highest intersection over union (IoU) of 96.03%, 84.56%, 70.11%, and 90.61%, respectively.

The rest of our article is detailed below. Section II is a review of relevant work. Section III provides details of the proposed method and each module. Section IV presents the experiment outcomes and the ablation experiments on various datasets. Last, we conclude our work in Sections V and VI.

II. Related Work

We introduce the related work in this section, which mainly includes traditional CD methods and deep learning-based CD models.

A. Traditional CD Methods

Traditional CD methods are usually performed in an unsupervised manner. It is split into three major steps: 1) image pretreatment; 2) difference map generation; and 3) difference map analysis to obtain the CD result. The third process is to identify changes often through clustering algorithms or thresholding algorithms. Threshold-based methods acquire feature maps though algebraic operations like difference operators, algebraic ratio operators, and logarithmic ratio operators [32]. However, such algorithms are highly dependent on expert

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experience for the threshold selection, which makes such methods prone to empirical errors and poor robustness [33].

In contrast to threshold-based methods, clustering-based algorithms are broadly applied in CD tasks. Those mainly comprise the c-means and the fuzzy c-means clustering (FCM) algorithms, but they cannot accurately catch spatial features and are poorly robust to noise [34]. Moreover, Markov random field (MRF) has been employed as an excellent tool to improve the clustering algorithm. Chatzis and Varvarigou [35] proposed the MRF-based FCM algorithm, which improves the detection accuracy, but it introduces artificially defined weighting parameters, which are not well adapted to complex tasks. Above algorithms depend on difference image production and they are sensitive to noise. Moreover, most clustering-based algorithms require different clustering strategies for different datasets resulting in poor robustness.

B. Deep Learning-Based CD Methods

Benefiting from the rapid development of computer vision and the appearance of large labeled datasets, many supervised deep learning methods are applied to RS image CD. For example, Long et al. [36] introduced an FCN, which employs full convolution operations for replacing the fully connected layer in traditional VGG model, which makes the FCN to be a fully end-to-end network model. FCN was originally used in image segmentation field, due to its excellent performance, which is also widely used in CD tasks. However, FCN lacks sufficient attention to the pixel-to-pixel relationship and suffers from a shortage of spatial consistency. Later, Ronneberger et al. [23] presented the UNet for medical image segmentation, where it introduces a strategy of skip connections, which permits the network to transmit contextual information from low resolution to high-resolution levels. Lv et al. [37] proposed Simple Multiscale UNet for CD, which embeds multiscale convolutional modules in UNet to capture target features of different sizes. Later, to capture more fine-grained features, Zhou et al. [38] proposed the UNet++, which allows multilevel feature fusion and reuse through dense skip connections. In the early stage, bi-temporal images are usually stitched together and later fed into network for the CD tasks. Li et al. [39] proposed the deep translation-based CD network (DTCDN), which uses deep translation to learn feature mapping between optical and synthetic aperture radar (SAR) RS images, followed by using improved UNet++ for CD tasks. Yet, the mentioned single-branch networks fail to exploit the bi-temporal images effectively, resulting in losing feature information of the original images. Consequently, LF strategy-based Siamese networks are developed and become the dominant framework in CD tasks.

The Siamese-based network was initially developed to solve the image matching problem, which comprises two branches with shared weights. Daudt et al. [25] first developed three Siamese-based networks for the CD task and obtained better detection results. Chen et al. [40] proposed a semantic decoupled representation learning (SDRL), the network is a Siamese architecture, which uses semantic masks embedded in contrastive self-supervised learning to guide the model distinguishing foreground and background features. Later, Fang et al. [41] used a Siamese network based on UNet++ to extract bi-temporal image features. Although the features are produced by weight-sharing encoders, the unchanged object features may vary significantly during this stage. Thus, the effective integration of the features remains an issue. To extract more discriminative feature information, Liu et al. [26] presented the SRCDNet, which takes a Siamese-based architecture and brings a convolutional block attention module (CBAM) module to enhance the valid information in the features. Subsequently, in order to better capture long-range semantic features, Chen et al. [42] proposed the bi-temporal image transformer (BIT), where transformer is introduced to simulate the spatial–temporal relationships of images. The bi-temporal images first go through the encoder for feature extraction, followed by the BIT for semantic tokens, then the tokens are fed back into the pixel-space and the feature maps are distilled by the transformer decoder. Later, Shi et al. [15] presented a deeply supervised attention metric-based network (DSAMNet), by using an attention mechanism and a deep supervision module for implementing global feature extraction. To focus on learning changed regions, Cheng et al. [43] proposed the deep learning networks that deliver improved separability (ISNet), which uses deformable convolution to implement an edge maximization strategy and uses channel and spatial attention to highlight location and semantic information. These networks all reinforce or focus on features in the decoding stage without further exploiting the rich semantic information in the encoding stage, which leads to the possibility of losing much spatial information in such approaches. In addition, to suppress the adverse effects of pseudo-variation, Peng et al. [44] proposed difference-enhancement dense-attention CNN (DDCNN), which extracts feature representations through dense skip connections and disparity enhancement modules. The difference enhancement module is a convolutional process after the direct difference of the image, which may cause the image to contain uncertainty regions, resulting in misled changed regions during the feature learning process. Inspired by feature fusion methods in the field of RS image [45], [46], [47], we proposed an efficient self-weighted spatial–temporal attention network (SSANet) in our previous work [48], which consists mainly of a fusion and a differential sub-network for extracting object and changed information, respectively. However, the method does not deeply explore the intrinsic connection between the difference and bi-temporal features.

Although the above methods have made great progress, however, the current methods have not yet conducted in-depth research on the extraction and integration of multiscale features. Moreover, how to design an effective difference-guided strategy to identify the variations is the key in the CD tasks. Based on the above problems, in this article, we design a novel CD network for generating accurate detection results.

III. METHOD

In this part, our proposed network is first described, and then the individual network modules will be described in detail.

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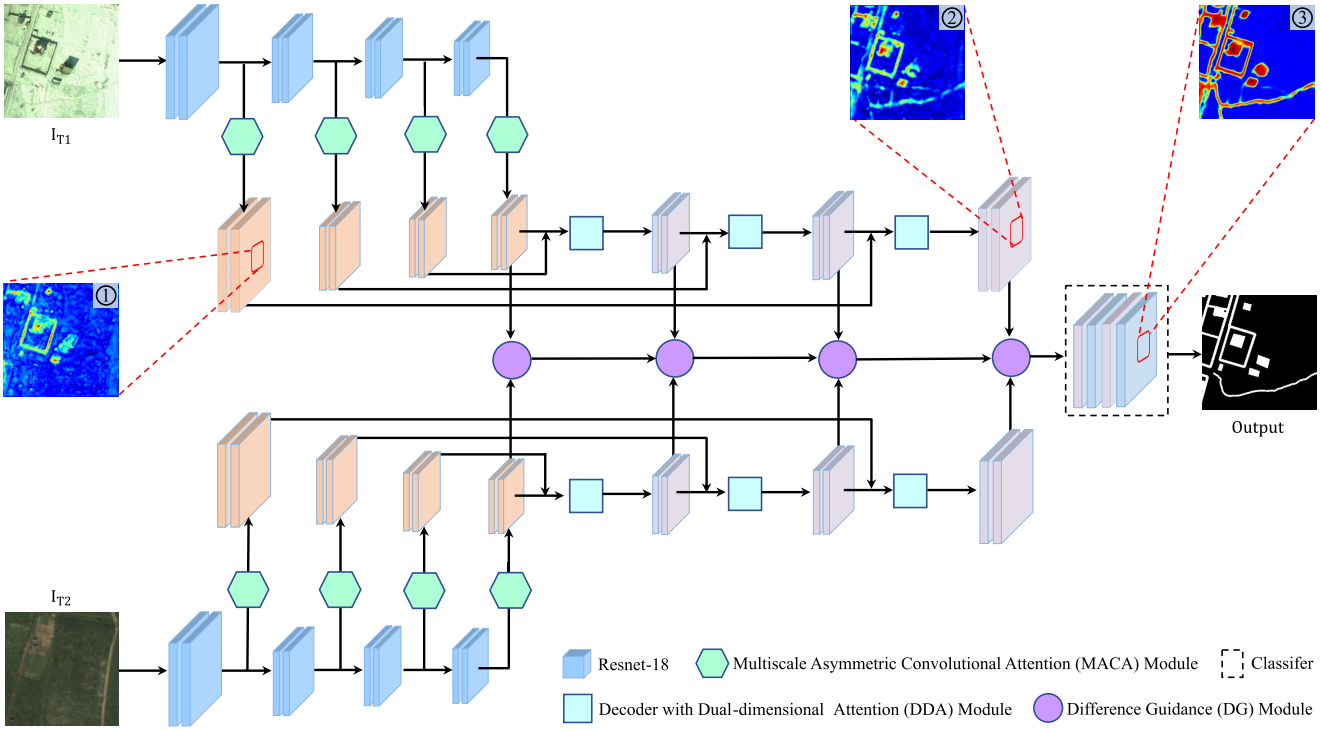


Fig. 1. Overall structure of MANet. ①–③ represent the visualization results of the feature maps after images passed through the MACA, DDA, and DG modules, respectively. Section III presents details of the different modules.

At the end, the training process of the model is presented, including the training procedure and the loss function.

A. Overview

The proposed network architecture is given in Fig. 1, where it is composed of a dual-branch encoder and decoder. The encoder mainly contains the feature extractor and MACA modules, which are used to learn the representative features of the image. The decoder consists of a DDA module and a DG module. The DDA module is used to achieve adaptive fusion of shallow and deep features. The DG module uses disparity information to guide feature learning and enhance the network's focus on changed regions.

Let I_{T1} and I_{T2} denote RS images of the same region acquired in different times $T1$ and $T2$, respectively, while M stands binary labels with variation annotations. The learning process of MANet can be described as follows.

1) First, images I_{T1} and I_{T2} are fed to the feature extractor, which consists of the popular weight-shared ResNet-18 [49]. After that a set of feature vectors are obtained for each image $F_n = \{f_{1n}, f_{2n}, f_{3n}, f_{4n}\}, n = T1, T2$.
2) Later, the feature vectors go to the MACA module to get the reinforced features $F'_n = \{f'_{1n}, f'_{2n}, f'_{3n}, f'_{4n}\}, n = T1, T2$, where the MACA module can extract more discriminative features.
3) Entering the decoder stage, F'_n enters the DDA module for adaptive fusion, and obtains the feature map $F''_n = \{f''_{1n}, f''_{2n}, f''_{3n}, f''_{4n}\}, n = T1, T2$. The image sizes of f''_{1n} to f''_{4n} are 16×16, 32×32, 64×64, and 128×128, respectively.
4) Finally, F''_n enters the DG module to get distance maps of different sizes $D_i = \{\text{dist1}, \text{dist2}, \text{dist3}, \text{dist4}\}$. After that, dist1, dist2, and dist3 are upsampled to size of dist4 and concatenated in the channel dimension to get D'_i. Next, D'_i enters the classifier for channel reduction, where the classifier consists of multiple layers of convolution, and then the prediction map D is obtained by upsampling. Finally, L_{bcl} loss function is used for model training.

B. MACA Module

RS images usually contain complex scenarios, where the changed regions are often hidden in the background of illumination changes, seasonal changes, etc. Current approaches use existing frameworks as encoders individually, such as ResNet and VGG [49], [50], which make it difficult to obtain the best feature representation. Therefore, to effectively extract representative information in the multiscale feature space, we design the MACA module. What is different is that we do not use the attention mechanism, but design a novel convolutional attention module. As Fig. 2 shows, MACA is composed of three parts: a 1×1 convolution for adding nonlinear features, a multibranch asymmetric convolution for capturing multiscale context, and a 1×1 convolution for channel modification. First, the input features enter the 1×1 convolution, which can be expressed as

$$F_0 = \text{Conv}_{1\times1}(X) \tag{1}$$

where X indicates the input feature graph and F_0 represents the feature graph obtained after 1×1 convolution. After that,

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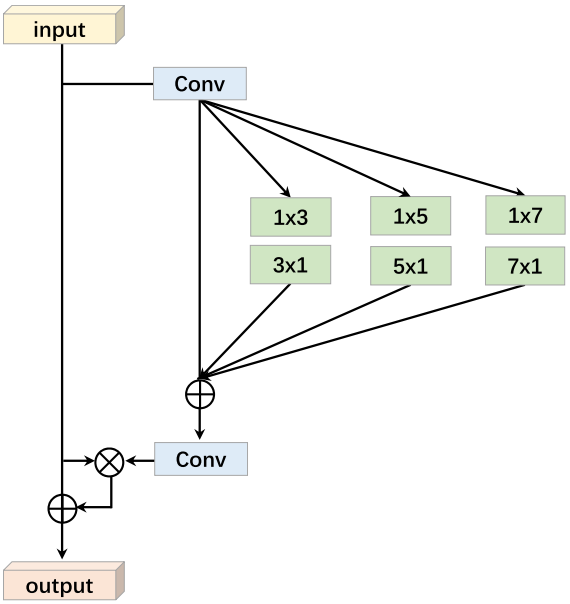


Fig. 2. Illustration of MACA.

the three feature maps are obtained by multibranch asymmetric convolution, the formula is as follows:

$$\begin{aligned} F_1 &= \text{Conv}_{3\times1}(\text{Conv}_{1\times3}(F_0)) \\ F_2 &= \text{Conv}_{5\times1}(\text{Conv}_{1\times5}(F_0)) \\ F_3 &= \text{Conv}_{7\times1}(\text{Conv}_{1\times7}(F_0)) \end{aligned} \tag{2}$$

where F_1, F_2, and F_3 represent the feature maps obtained by asymmetric convolution of different branches, with kernel sizes of 3, 5, and 7, respectively. There are two reasons why we choose asymmetric convolution. First, asymmetric convolution can maintain better feature extraction ability for image edges and strip objects. Second, the parameter number of asymmetric convolution is smaller than normal convolution, which contributes to the reduction of network parameters. Later, the different feature maps are summed to obtain the attention weights. It is worth mentioning that if we use only asymmetric convolution as weights, the closeness between features will be reduced [51]. Therefore, to counteract for the adverse effects caused by asymmetric convolution, we use the feature map F_0 after 1×1 convolution as part of the weights as well. The output after 1×1 convolution is used as the final attention to weight the input features, after which it is summed with the input feature map to gain output map, with the following equation:

$$\begin{aligned} F' &= F_0 + F_1 + F_2 + F_3 \\ \text{Att} &= \text{Conv}_{1\times1}(F') \\ \text{Out} &= \text{Att} \times X + X \end{aligned} \tag{3}$$

where F' denotes the feature after summing the outputs of different branches. Att denotes the attention weights and Out denotes the output. As the above analysis shows, the presented MACA module can effectively enhance the representation of multitemporal information and also enable the model to achieve optimal feature representation with fewer parameters.

C. DDA Module

The encoder stage produces low-level and deep-level semantic feature maps, and the decoder is applied to feature upsampling and generating prediction maps. In general, low-level semantic features include rich details, and high-level semantic features are useful for discerning changed regions. Therefore, we propose the DDA module in the upsampling phase, which is used to achieve efficient fusion of low-level and deep-level features. The upsampling module is shown in Fig. 3(a). The deep features are upsampled so that the features at different levels have the same size. Later, as shown in Fig. 3(b), these features are linked together in the channel dimension and enter the DDA module for feature selection in the feature and convolutional kernel dimensions, thus realizing the upsampling process of the network.

In Fig. 3(b), the fused feature map F is input to the DDA module. Adaptive average pooling is first applied to squeeze global space information, and then the fully connected layer is used for channel reduction, which can be expressed as

$$F_d = \text{Relu}(\text{FC}(\text{Avg}(F)) \tag{4}$$

where Avg denotes the adaptive average pooling operation, FC denotes the fully connected layer, and Relu denotes the rectified linear unit (ReLU) activation function. After the operation, the feature vector F_d is obtained. Later, to achieve parallel attention in the feature dimension and the convolution dimension, F_d will enter two branches. In the convolutional attention branch, F_d passes through a fully connected layer to generate a specific channel, which we set to 4 in this article. After that, SoftMax function is employed to calculate the normalized attention weights for four convolutional kernels, and we add a weight constraint of 1 to enhance model learning [52]. The process is as follows:

$$\begin{aligned} N_i &= \text{FC}(F_d) \\ \alpha_i &= \text{SoftMax}(N_i), \quad \text{for} \quad i = 1, \ldots, 4 \\ &\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 1 \end{aligned} \tag{5}$$

where N_i represents the features obtained after a fully connected layer. It is notable that the subscript i of N denotes the channel number, which is set to 4. α_i is the weight generated by SoftMax function, and it includes the channel weights and compressed spatial information. Here, we use four convolutional kernels to generate the final convolutional kernels, of which the parameters are initialized by the Kaiming initializer method [53]. The formula is as follows:

$$\begin{aligned} \text{Con}_i' &= \text{Con}_i \times \alpha_i, \quad \text{for} \quad i = 1, \ldots, 4 \\ \text{Conv} &= \text{Con}_1' + \text{Con}_2' + \text{Con}_3' + \text{Con}_4' \end{aligned} \tag{6}$$

where Con_i denotes the Kaiming initialized convolution kernel, and after multiplying with the weights, the attentional convolution kernel Con_i' is obtained. After that, the weighted convolution kernels are aggregated to obtain the final convolution Conv.

Meanwhile, in the feature attention branch, F_d goes through the fully connected layer for channel recovery. The purpose of adding a fully connected layer in the module is to combine the linear information between channels more efficiently. After

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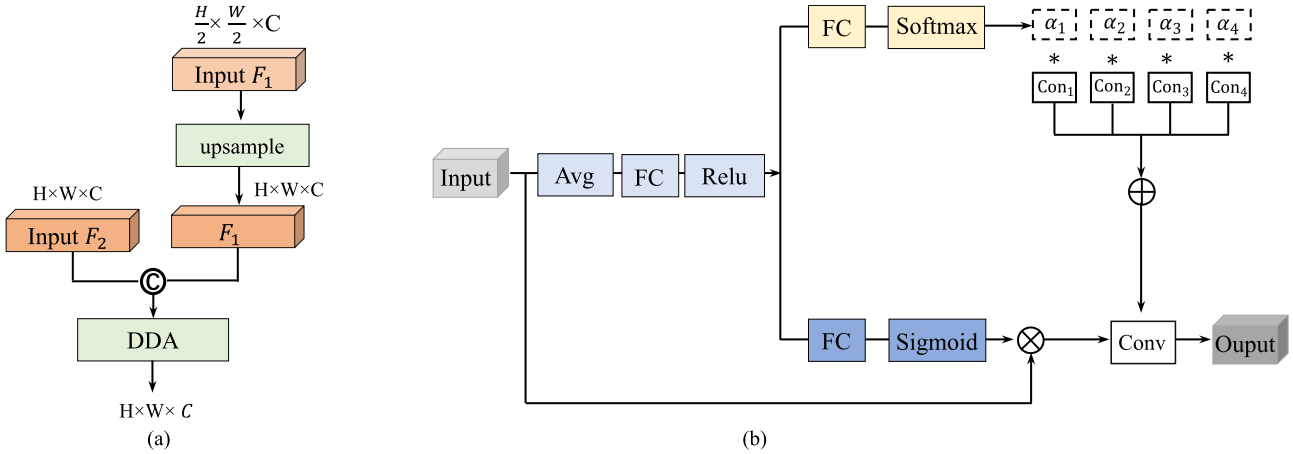


Fig. 3. Illustration of DDA. (a) Decoder block. (b) DDA module.

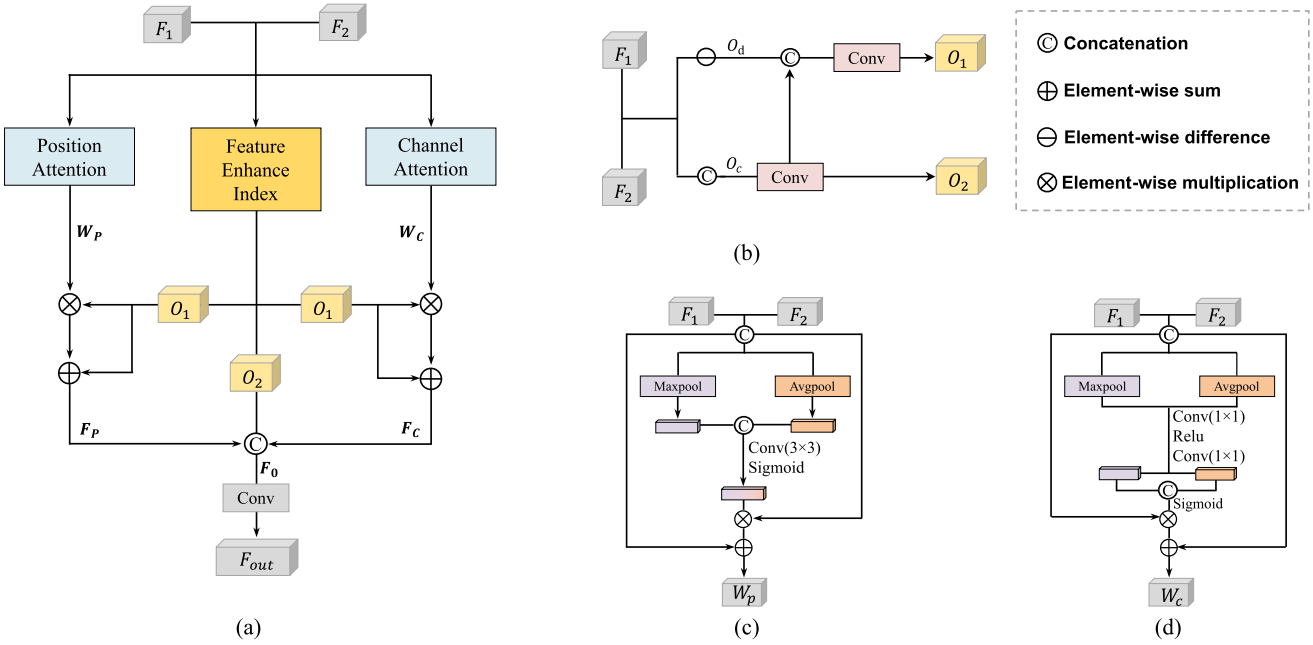


Fig. 4. Illustration of DG. (a) DG module. (b) DIM. (c) PAM. (d) CAM.

that, the channel dimension weights are obtained after the Sigmoid activation function, which can assign weights to the channels after channel interaction. Later, the weights are multiplied by the input feature maps with the following equation:

$$W_{\text{att}} = \text{Sigmoid}(\text{FC}(F_d))$$
$$F'' = W_{\text{att}} \times F \tag{7}$$

where W_{att} denotes the generated feature channel weights and F'' denotes the feature map after attention. With the feature attention F'' and convolutional attention Conv are generated in the two branches, F'' is sent into Conv to get the final Output. The weight formula is as follows:

$$\text{Output} = \text{Conv}(F''). \tag{8}$$

As shown above, DDA module learns two types of attention in two dimensions in a parallel manner, and it implements multidimensional attention aggregation in feature dimension and convolutional kernel dimension. Therefore, DDA can significantly enhance the feature discrimination capability of the network to achieve adaptive attention for both shallow and deep features.

D. DG Module

To strengthen the learning of variation between upsampled parallel features, we introduce the DG module. The DG module contains three sub-modules, namely difference index module (DIM), position attention module (PAM), and channel attention module (CAM). The global structure and specific details of the DG module are shown in Fig. 4.

Through DIM, network learns the difference features O_1 and fusion features O_2 at different levels. PAM and CAM are used to obtain the position-attention features W_p and channel-attention features W_c. Based on W_p and W_c, we selectively reinforce and attenuate differential feature O_1 in spatial and channel dimension, respectively, to realize the attention

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to the changed regions and suppress the effect of pseudo-variation. Thus, the difference features F_p guided by the PAM and the difference features F_c guided by the CAM are obtained, which are calculated as follows:

$$\begin{aligned} F_p &= (W_p \times O_1) + O_1 \\ F_c &= (W_c \times O_1) + O_1 \end{aligned} \tag{9}$$

where $+$ indicates element-wise addition operation and $\times$ represents element-wise multiplication operation. After that, F_p, F_c, and O_2 (i.e., generated by DIM) are concatenated to produce F_0. The final F_{out} is acquired after convolution operation, batch normalization, and activation function. The equation is described as follows:

$$\begin{aligned} F_0 &= \text{Concat}(F_p, F_c, O_2) \\ F_{\text{out}} &= \text{Relu}(\text{BN}(\text{Conv}_{3\times3}(\text{Conv}_{1\times1}(F_0)))) \end{aligned} \tag{10}$$

where $\text{Conv}_{1\times1}$ does not change the feature channels, $\text{Conv}_{3\times3}$ is used to achieve channel reduction, and the final output dimension is identical to the input dimension. BN denotes the batch normalization layer.

1) DIM: The DIM is to get the variation between the same-scale features F_1 and F_2. DIM is split into two branches, namely the absolute differential branch and the concatenation branch. The absolute differential branch is first employed to generate the difference map O_d, followed by concatenation with the feature map O_c from the concatenation branch as the final difference feature O_1. The purpose of the above operation is to mitigate the influence of noise caused by the direct use of O_1, thus it can adaptively learn the meaningful changed regions. The description is as follows:

$$\begin{aligned} O_d &= |F_1 - F_2| \\ O_c &= \text{Concat}(F_1, F_2) \\ O_1 &= \text{Conv}_{1\times1}(\text{Concat}(O_d, \text{Conv}_{1\times1}(O_c))) \end{aligned} \tag{11}$$

where the two $\text{Conv}_{1\times1}$ are used to reduce the channel to half of its original size, making the output channel identical to the input feature. In concatenation branch, O_1 goes through $\text{Conv}_{1\times1}$ for channel changes to get the final fusion output O_2, and the equation is as follows:

$$O_2 = \text{Conv}_{1\times1}(O_1). \tag{12}$$

2) PAM: For the input features $F_1, F_2 \in R^{C\times H\times W}$, we first perform channel dimension concatenation on the input features. The description is as follows:

$$S = \text{Concat}(F_1, F_2) \tag{13}$$

where Concat denotes the concatenation operation and S denotes the features obtained after concatenation. After that, max pooling and average pooling are applied to squeeze the input channel refinement feature S, and two $1 \times H \times W$ matrices are obtained. Finally, the two matrices are concatenated and fed into a convolutional layer with kernel size of 3. The Sigmoid function is applied to get the final spatial attention map M_p. The formula can be denoted as follows:

$$M_p = \sigma(\text{Conv}_{3\times3}(\text{Concat}(\text{Max}(S), \text{Avg}(S)))) \tag{14}$$

where Max denotes the max pooling operation, while σ denotes the Sigmoid activation function. Later, the spatial attention weights M_p are multiplied by S and the location-enhancing features W_p are generated using the residual connection method. The formula is as follows:

$$W_p = M_p \times S + S. \tag{15}$$

After the above operations, the spatial features related to the changed regions are enhanced and the irrelevant parts are weakened.

3) CAM: Similar to the PAM, the CAM aims to enhance the feature channels associated with the changes. The specific calculations are as follows:

$$\begin{aligned} F_c &= \text{Concat}(F_1, F_2) \\ M_c &= \sigma(\text{MLP}(\text{Max}(F_c)), \text{MLP}(\text{Avg}(F_c))) \\ W_c &= M_c \times F_c + F_c \end{aligned} \tag{16}$$

where F_c denotes the merged feature, based on it, max pooling and average pooling are used to produce two matrices of size $C \times 1 \times 1$. Then the multilayer perception (MLP) module is used to assign weights to each channel, where the MLP contains two 1×1 convolutional layers for channel reduction and recovery, respectively, which are connected by the Relu function. Afterward, the channel attention weight M_c is obtained after the Sigmoid function. Finally, the original splicing features are multiplied with channel attention weights and then summed to obtain the channel attention features W_c.

E. Loss Function

In bi-temporal images, the ratio of unchanged regions is usually much larger than the ratio of changed regions, which makes the network sensitive to imbalance problems during the training process. To alleviate this problem, we employ batch balance contrast loss (BCL) [54] in our experiments. The purpose is to assign corresponding weights to different loss terms to reach a balance. The loss function equation is given below

$$\begin{aligned} L(D^*, M^*) = &\frac{1}{2}\frac{1}{p_u}\sum_{b,i,j}\left(1 - M^*_{b,i,j}\right)D^*_{b,i,j} \\ &+ \frac{1}{2}\frac{1}{p_c}\sum_{b,i,j} M^*_{b,i,j}\text{Max}\left(0, m - D^*_{b,i,j}\right) \end{aligned} \tag{17}$$

where D^* and M^* denote the prediction map and binary labels, respectively, where 1 means changed and 0 means unchanged. m stands for boundary value that limits the range of changing pixels, which in our experiments is set to 1.5. p_u denotes the unchanged pixel numbers and p_c stands for the changed pixel counts. They can be calculated by summing the labels of the corresponding categories

$$p_u = \sum_{b,i,j} 1 - M^*_{b,i,j}, \quad p_c = \sum_{b,i,j} M^*_{b,i,j}. \tag{18}$$

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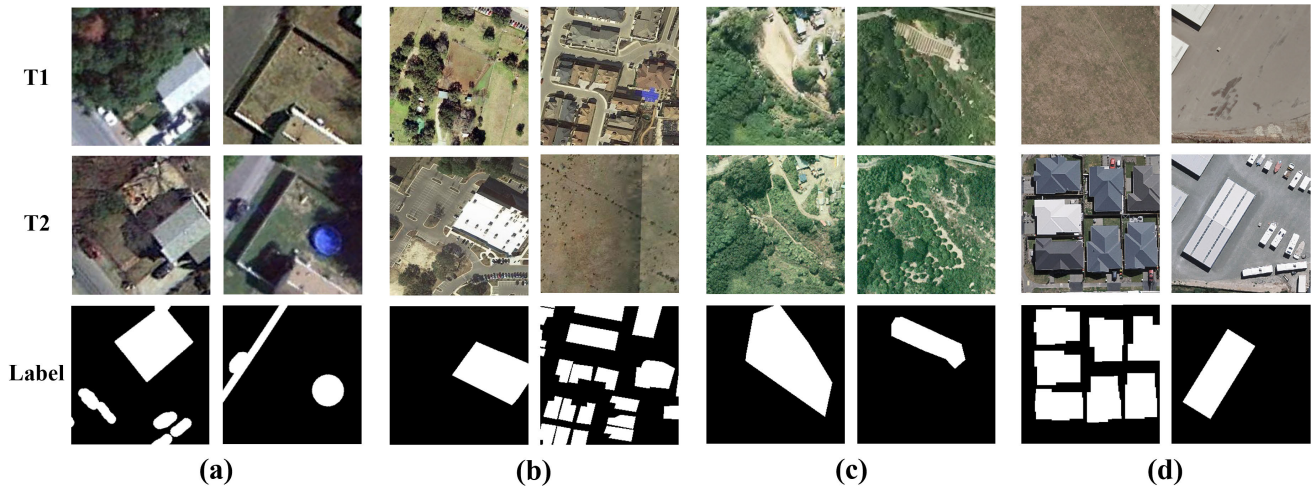


Fig. 5. Bi-temporal images from four datasets. (a) CDD. (b) LEVIR-CD. (c) SYSU-CD. (d) BCDD.

IV. Experiments

The experiments are implemented on four popular public CD datasets, CDD [55], LEVIR-CD [56], SYSU-CD [15], and BCDD [57]. To verify the performance of the presented method, we use ten advanced networks as comparative methods. In addition, ablation experiments are executed to demonstrate the effectiveness of each individual module. Finally, the visualization of the feature maps for each module is discussed.

A. Datasets

1) CDD: It contains 16 000 pairs of seasonal variation RS images, which include seasonal, building, and vehicle variations. Each image pair is 256 × 256 pixels and resolution from 0.03 to 1 m. In the experiment, 10 000 image pairs are allocated for training, 3000 pairs as validation, and the other 3000 pairs are assigned for test. To avoid overfitting, we perform a data enhancement operation. Some samples in CDD are shown in Fig. 5(a).

2) LEVIR-CD: The LEVIR-CD dataset contains 637 pairs of very-high resolution Google Earth images, each of which has a size of 1024 × 1024 pixels with 0.5 m resolution. This dataset comes from 20 different urban areas in USA, and it contains variation objects such as villas, high-rise buildings, and warehouses. In our experiment, we divide the image into 256 × 256 size image blocks and the division process has no duplicate regions. The ratios of training, verification and testing are 7:1:2, respectively. To enhance data diversity, we execute data enhancement operations. Images in LEVIR-CD are shown in Fig. 5(b).

3) SYSU-CD: The SYSU-CD dataset contains 20 000 pairs of RS images, with ground objects of high-rise buildings, harbor, and suburban variations from Hong Kong area. Each image pair consists of 256 × 256 pixels in size with 0.5 m resolutions. For experiments, the ratios of trained, validated, and tested are 6:2:2, respectively. The main types of SYSU-CD are shown in Fig. 5(c).

4) BCDD: This dataset comprises pairs of 0.2 m resolution images with dimensions of 32 507 × 15 354 pixels. These images depict building changes that occurred before and after an earthquake, encompassing various types of changes such as reconstruction and renewal. To facilitate model training and testing, we partition these images into 7434 non-overlapping patches, each with dimensions of 256 × 256 pixels. These patches are randomly assigned to training, validation, and testing sets in an 8:1:1 ratio. Some samples are shown in Fig. 5(d).

B. Comparative Methods

To substantiate the validity of the proposed network, ten state-of-the-art CD approaches are used for comparison. The above methods are briefly outlined as follows.

1) Fully Convolutional-EF (FC-EF) [25]: It is based on an improvement of the UNet, which uses the EF strategy, where bi-temporal image concatenation is used as input and the encoder is applied to obtain the prediction results.

2) Fully Convolutional-Siamese-Concatenation (FC-Siam-conc) [25]: This network is based on another variant of the UNet, which uses the LF strategy. The network uses Siamese architecture, and features of the encoder are concatenated and then skip connection is made into the decoder.

3) Fully Convolutional-Siamese-Difference (FC-Siam-diff) [25]: As another variant of UNet, it makes difference between the features from two shared-weight encoders and then makes a skip connection with the decoder for feature fusion.

4) Super-Resolution-Based CD Network [26]: A metric-based network that uses a CBAM attention mechanism to strengthen channel and spatial feature maps.

5) BIT Network (BITNet) [42]: BITNet introduces the transformer mechanism to the Siamese-based network, which allows efficient and effective context modeling in the spatial–temporal domain.

6) Difference-Enhancement Dense-Attention CNN [44]: A UNet-based structure that introduces dense skip connections to aggregate shallow features and deep features. Furthermore, a difference module is added to guide the final result.

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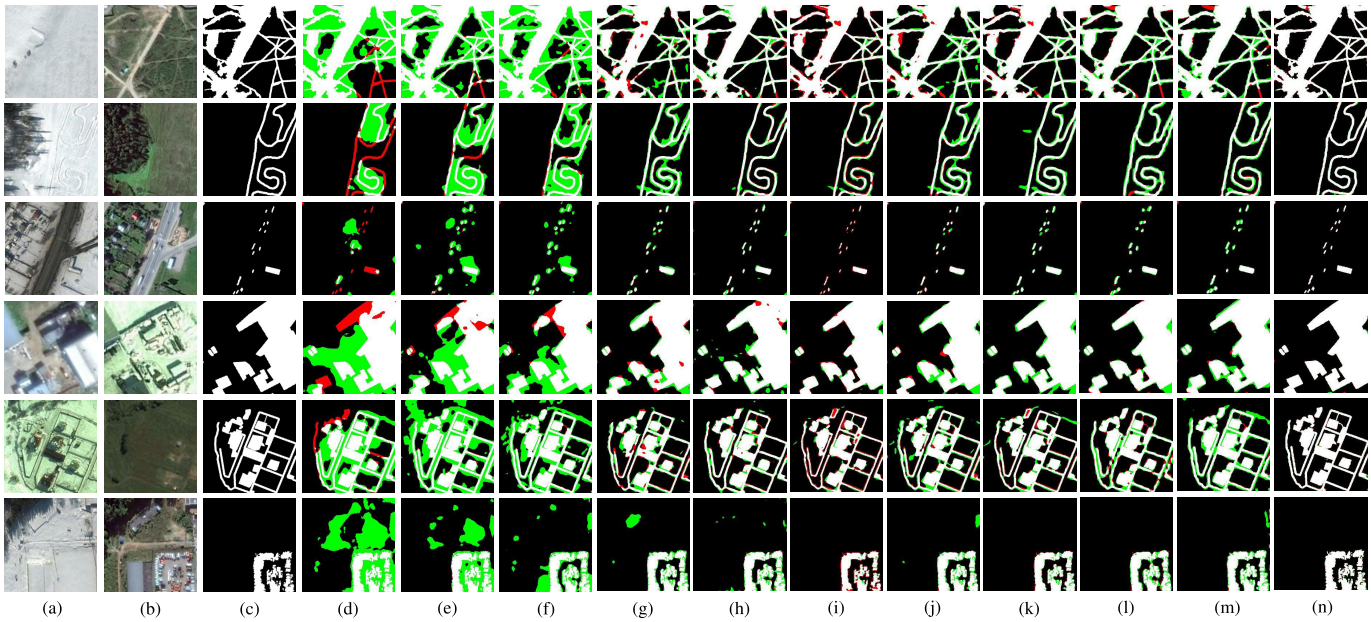


Fig. 6. Comparisons on the CDD dataset. (a) $T1$ instance. (b) $T2$ instance. (c) Label. (d)–(n) FC-EF, FC-Siam-diff, FC-Siam-conc, SRCDNet, BITNet, DDCNN, DSMANet, ISNet, DETNet, UGSNet, and our MANet; colors: black for TN (i.e., "unchanged"), white for TP (i.e., "changed"), green for FP, and red for FN.

7) Deeply Supervised Attention Metric-Based Network [15]: A deep-supervised Siamese network, where the deep supervised module is used to reinforce the learning capability of the feature extractor.

8) Deep Learning Networks That Deliver Improved Separability (ISNet) [43]: A Siamese network that introduces spatial and channel attention to highlight both semantic and location information. Meanwhile, deformable convolution is applied to perform feature refinement, thus highlighting the change boundary.

9) Deep Edge-Guided and Transferable Super-Resolution Mapping Network (DETNet) [58]: It uses progressively edge-guided super-resolution reconstruction block to reconstruct feature details and adaptively localize edges at multiple scales.

10) FCN for Fine-Grained UGS Mapping (UGSNet) [59]: A network that introduces an enhanced coordinate attention module and a point head module. In our experiments, we use UGSNet without discriminator as comparison due to the task requirements.

C. Experimental Setup

1) Implementation Details: We employ the pytorch architecture to implement our network, using an NVIDIA GeForce GTX3090 GPU with 24G of graphic memory for training and testing. Additionally, we work with random flipping and cropping to achieve data enhancement. During training, the Adam gradient descent method is used to optimize the model, where the learning rate, batchsize, and momentum are setup as 0.000125, 10, and 0.99, respectively.

2) Evaluation Metrics: For comparing the different method performance, we employ four metrics to evaluate the correlation between predicted change map and labels, which include Precision (Pr), $F1$-score ($F1$), Recall (Rc), and IoU. Here, Precision stands for the false positive ratio, and Recall represents the leakage ratio. $F1$ refers to the harmonic mean from Precision and Recall, and IoU refers to the overlapping area on the test map and the label in relation to its union. These indicators can be expressed as follows:

$$\mathrm{Pr} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{19}$$

$$\mathrm{Rc} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{20}$$

$$F1 = \frac{2\mathrm{Pr} \cdot \mathrm{Rc}}{\mathrm{Pr} + \mathrm{Rc}} \tag{21}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{22}$$

D. Comparing Experiments

1) Comparisons on CDD Dataset: The experimental outcomes of different approaches on the CDD dataset are displayed in Fig. 6, and the quantitative analyses are listed in Table I. The inclusion of complex variation types such as roadways, trees, and buildings makes the detection results susceptible to noise and pseudo-change, which can lead to false detections. As seen in the figure, our network achieves better detection results with fewer misdetections and omissions. The image in the first row of Fig. 6 contains complex variations, such as road, seasonal, and illumination changes. Due to the insufficient feature extraction capability of the network, there are obvious false detections in Fig. 6(d)–(g), and discontinuities in the changed regions will be found in Fig. 6(h)–(m). Compare to other methods, our network is closer to the real labels in processing such complex change scenarios. The fifth row of images in Fig. 6 contains variations of the types of large and strip-like buildings. Fig. 6(d)–(m) is incomplete for the preservation of the change boundary, and our network has better feature extraction ability for strip-like or large buildings because it contains MACA module. Therefore, our network

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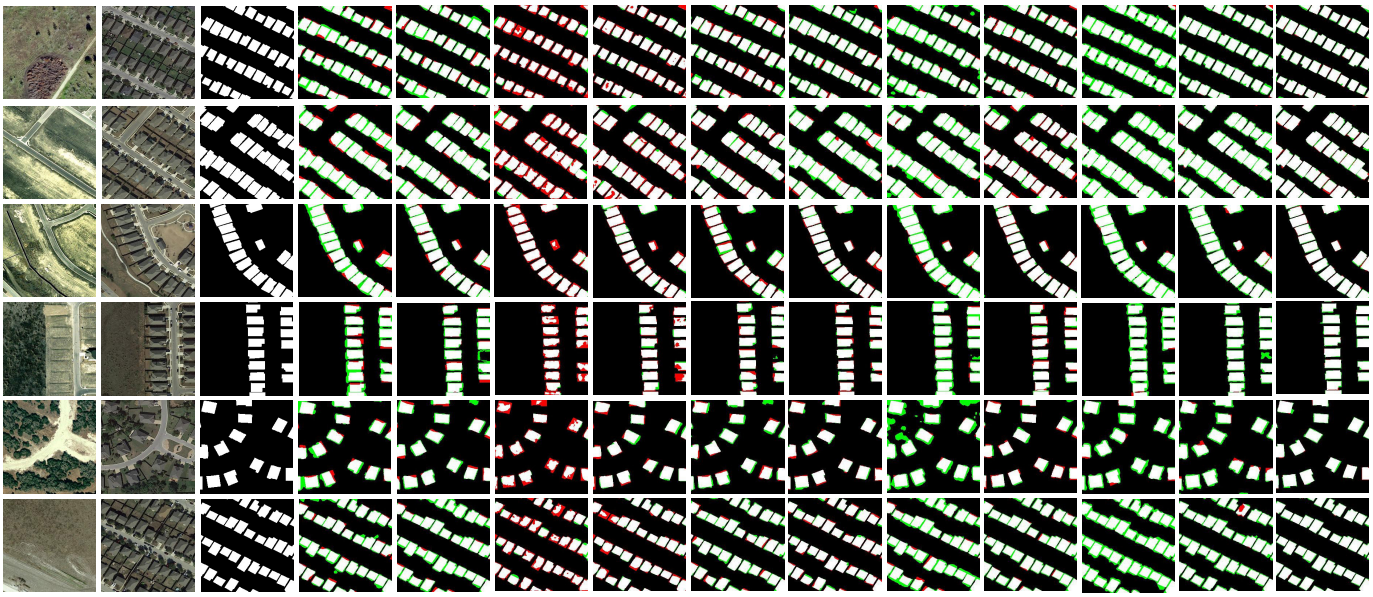


Fig. 7. Comparisons on the LEVIR-CD dataset. (a) $T1$ instance. (b) $T2$ instance. (c) Label. (d)–(n) FC-EF, FC-Siam-diff, FC-Siam-conc, SRCDNet, BITNet, DDCNN, DSMANet, ISNet, DETNet, UGSNet, and our MANet; colors: black for TN (i.e., "unchanged"), white for TP (i.e., "changed"), green for FP, and red for FN.

TABLE I

COMPARATIVE EXPERIMENTS RESULTS ON CDD DATASET. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN BLUE AND RED, RESPECTIVELY

Methods	Pr(%)	Rc(%)	F1(%)	IoU(%)
FC-EF	84.99	64.14	73.11	57.62
FC-Siam-diff	82.01	83.89	82.94	70.85
FC-Siam-conc	86.66	81.73	84.12	72.60
SRCDNet	94.58	91.40	92.96	86.85
BITNet	93.17	95.16	94.15	88.95
DDCNN	97.63	91.24	94.33	90.65
DSAMNet	95.96	91.77	93.81	88.35
ISNet	96.85	93.53	95.16	90.78
DETNet	95.28	94.97	95.12	90.71
UGSNet	94.46	94.32	94.39	89.38
Ours	98.20	97.74	97.97	96.03

TABLE II

COMPARATIVE EXPERIMENTS RESULTS ON LEVIR-CD DATASET. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN BLUE AND RED, RESPECTIVELY

Methods	Pr(%)	Rc(%)	F1(%)	IoU(%)
FC-EF	84.67	81.03	83.68	71.94
FC-Siam-diff	89.11	83.02	85.96	75.37
FC-Siam-conc	92.64	76.89	84.03	72.46
SRCDNet	90.73	81.55	85.90	75.29
BITNet	89.93	89.45	89.69	81.31
DDCNN	93.16	87.34	90.16	82.08
DSAMNet	80.19	89.06	84.39	73.01
ISNet	93.38	87.51	90.35	82.40
DETNet	89.16	87.31	88.23	78.93
UGSNet	90.39	89.28	89.83	81.55
Ours	91.23	92.04	91.64	84.56

achieves more fine-grained detection result. To further verify the superior performance of proposed network, the outcomes of the quantitative evaluation are provided in Table I. It shows that our network obtains the highest Pr, Rc, $F1$, and IoU, which are 1.35%, 4.21%, 2.81%, and 5.25% higher than the state-of-the-art method, respectively.

2) Comparisons on LEVIR-CD Dataset: The outcomes of the LEVIR-CD dataset are displayed in Fig. 7, and the quantitative result is shown in Table II, where bold indicates the best value. On the one hand, there are false detections in dense small building changed regions due to illumination and registration errors, as demonstrated in the first row of Fig. 7(d)–(h). Nevertheless, our approach generally achieves a better boundary accuracy with no significant sticking. On the other hand, for large building variations, the smoothness of the boundary and the error detection are decisive factors in judging results. The fifth row of Fig. 7(d)–(m) shows different degrees of these problems. In the third row, Fig. 7(f)–(i) shows more obvious missing detections and edges, while Fig. 7(d), (e), and (j)–(m) shows false detections. Our network demonstrates optimal performance in maintaining building integrity. To further illustrate the superiority of the proposed approach, the quantitative assessment outcomes are exhibited in Table II. Our network clearly outperforms other methods,

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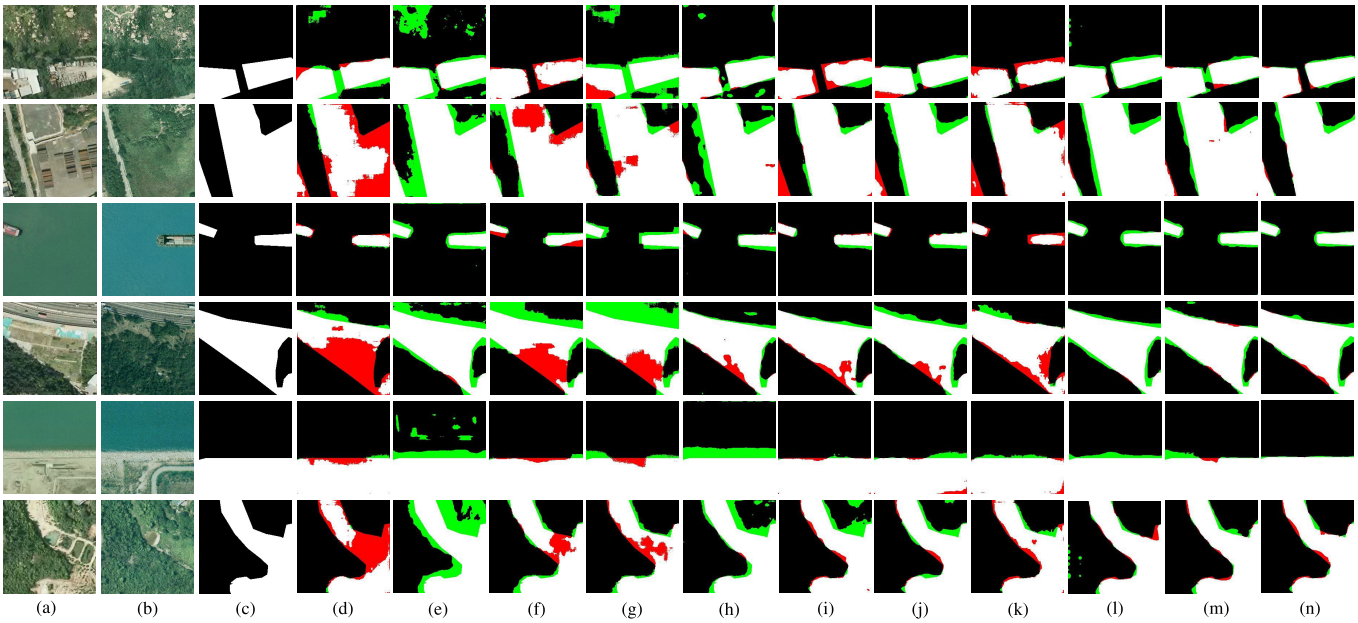


Fig. 8. Comparisons on the SYSU-CD dataset. (a) $T1$ instance. (b) $T2$ instance. (c) Label. (d)–(n) FC-EF, FC-Siam-diff, FC-Siam-conc, SRCDNet, BITNet, DDCNN, DSMANet, ISNet, DETNet, UGSNet, and our MANet; colors: black for TN (i.e., "unchanged"), white for TP (i.e., "changed"), green for FP, and red for FN.

TABLE III
COMPARATIVE EXPERIMENTS RESULTS ON SYSU-CD DATASET. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN BLUE AND RED, RESPECTIVELY

Methods	**Pr(%)**	**Rc(%)**	**F1(%)**	**IoU(%)**
FC-EF	81.61	69.63	75.14	60.18
FC-Siam-diff	63.58	87.08	73.50	58.11
FC-Siam-conc	82.36	72.20	76.95	62.53
SRCDNet	78.64	73.57	76.02	61.32
BITNet	68.78	88.45	77.39	63.12
DDCNN	79.71	78.54	79.12	65.34
DSAMNet	78.80	77.68	78.23	64.25
ISNet	81.02	76.06	78.46	64.57
DETNet	77.44	83.08	80.16	66.89
UGSNet	80.54	81.43	80.98	68.05
Ours	80.63	84.30	82.42	70.11

where it achieves the highest Rc, $F1$, and IoU, with 92.04%, 91.64%, and 84.56%, respectively. This shows our network has an efficient feature extraction capability for both small and large change objects, thus maintaining a good change edge and internal compactness.

3) Comparisons on SYSU-CD Dataset: The outcomes of the SYSU-CD dataset are displayed in Fig. 8. The analysis is shown in Table III, where the blue font indicates the best results. As viewed in the second row of Fig. 8, although Fig. 8(d)–(h) can extract significant changes, they still have more false detections. Fig. 8(i)–(m) can capture relatively complete changed regions, including road and building changes, but they still have the problem of blurred boundaries. In general, our network obtains better detection results while maintaining the complete changed regions. In addition, they can better distinguish between pseudo-variation and true-variation, such as the first and the fifth row of Fig. 8. To further demonstrate the superiority of our approach, the quantitative results are shown in Table III. As the criteria indicates, our network performance achieves the most accurate results with IoU 4.77% higher than the state-of-the-art method, which indicates our network has better representation in maintaining information about the boundaries of changes. FC-Siam-conc has the highest Pr of 82.36%, but its Rc is only 72.20%, which means the network prefers to judge the changed regions as unchanged regions. In contrast, BITNet has the highest Rc of 88.45%, but its Pr is only 68.78%, which indicates that the model is more inclined to misclassify unchanged regions as changed regions. Fig. 8 also confirms the above analysis, the high Pr and low Rc of the FC-Siam-conc cause significant missed detection of test results. The high Rc and low Pr of BITNet lead to serious false detections. Proposed method obtains the highest values of $F1$ and IoU because of better balance between Pr and Rc. Therefore, our method can accurately capture the changed regions while effectively controlling false and missed detections. This demonstrates the proposed network offers more reliable detection performance. In addition, the metrics for each method are lower than the CDD dataset, and it is due to SYSU-CD dataset includes much more complex types of variations.

4) Comparisons on BCDD Dataset: Fig. 9 shows the experimental results for the BCDD dataset and Table IV shows the analysis results. As seen from experiments, our method achieves the best detection results. In the second row, Fig. 9(d)–(m) can detect the overall changes, but they still have many false detections and omissions, and MANet represents

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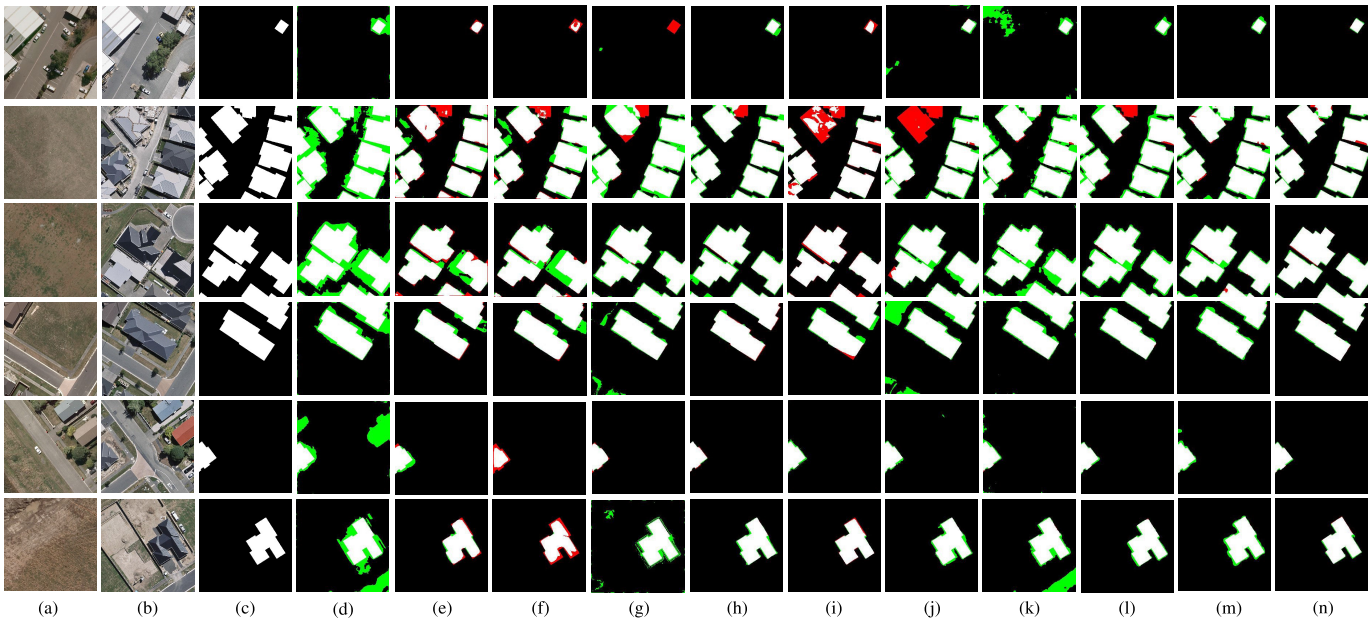


Fig. 9. Comparisons on the BCDD dataset. (a) $T1$ instance. (b) $T2$ instance. (c) Label. (d)–(n) FC-EF, FC-Siam-diff, FC-Siam-conc, SRCDNet, BITNet, DDCNN, DSMANet, ISNet, DETNet, UGSNet, and our MANet; colors: black for TN (i.e., "unchanged"), white for TP (i.e., "changed"), green for FP, and red for FN.

TABLE IV
COMPARATIVE EXPERIMENTS RESULTS ON BCDD DATASET. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN BLUE AND RED, RESPECTIVELY

Methods	**Pr(%)**	**Rc(%)**	**F1(%)**	**IoU(%)**
FC-EF	86.40	69.83	77.24	62.92
FC-Siam-diff	78.41	90.49	84.02	72.45
FC-Siam-conc	86.55	78.64	82.41	70.08
SRCDNet	91.46	89.62	90.53	82.71
BITNet	94.57	91.26	92.88	86.72
DDCNN	97.32	86.28	91.47	84.28
DSAMNet	85.66	74.99	79.97	66.63
ISNet	88.95	86.02	87.46	77.71
DETNet	90.47	91.46	90.06	83.42
UGSNet	87.81	91.71	89.72	81.36
Ours	94.63	95.01	94.82	90.16

more accurate results. In the quantitative analysis, MANet achieved the highest IoU and $F1$, which are 90.16% and 94.82%, respectively. This indicates the proposed method is able to extract fine-grained features efficiently, thus producing better detection results.

E. Ablation Experiments

1) Ablation Experiments of Different Modules: To verify the primary module's effectiveness (MACA, DDA, DG), we perform a series of ablation experiments on the CDD, LEVIR-CD, SYSU-CD, and BCDD datasets. During ablation analysis, the $F1$ and IoU are considered as a comprehensive evaluation index. In addition, we maintain the same experimental details and parameters to the comparison experiment. The baseline is a variation of MANet, which only retains the Siamese encoder–decoder structure, and skip connections. The experimental outcomes are exhibited in Table V.

As shown in the table, baseline model achieves the lowest detection results on all four datasets. With the insertion of MACA, DDA, and DG modules, network performance has gradually improved. The "Baseline + DG" model shows the greatest improvement in four datasets, with IoU improving by 4.84% for CDD, 3.21% for SYSU-CD, 3.46% for LEVIR-CD, and 2.82% for BCDD. With the combination of three modules, "Baseline + MACA + DDA + DG" model, namely MANet, outperforms other networks in ablation studies. Its $F1$ and IoU increase by 3.49% and 6.48% for the CDD dataset, 4.60% and 6.41% for the SYSU-CD dataset, 3.21% and 5.30% for the LEVIR-CD dataset, 3.08% and 5.43% for the BCDD dataset, respectively. As a result, the quantitative analysis of the ablation experiments demonstrated not only the individual effectiveness of the MACA, DDA, and DG modules, but also their excellent combined synergistic properties.

2) Ablation Experiments of DG Module: Table VI demonstrates the validity of the submodules in DG block. In our experiment, we keep other parts of MANet unchanged and remove the PAM, CAM, and DIM from the DG module as the baseline. It can be observed that baseline achieves the lowest performance on all four datasets, due to its insufficient use of variance information. With the gradual adoption of PAM, CAM, and DIM submodules, the network performance has been gradually improved. Compared to baseline, the network that introduces three submodules, i.e., MANet, improves its IoU by 1.34% on the CDD dataset, 1.42% on the LEVIR-CD dataset, 1.29% on the SYSU-CD dataset, and 1.30% on the BCDD dataset. The experimental outcomes demonstrate

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TABLE V
ABLATION STUDY ON FOUR DATASETS. VALUES IN BOLD FONT ARE THE BEST. × MEANS NO COMBINATION. ✓INDICATES THE MODULE HAS BEEN ADDED TO THE BACKBONE

Module			CDD		LEVIR-CD		SYSU-CD		BCDD	
MACA	DDA	DG	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
×	×	×	94.48	89.55	88.43	79.26	77.82	63.70	91.74	84.73
✓	×	×	97.06	94.30	90.42	82.53	79.22	65.60	93.04	86.98
×	✓	×	96.67	93.75	90.23	82.21	78.89	65.15	92.13	85.41
×	×	✓	97.11	94.39	90.54	82.72	80.17	66.91	93.36	87.55
✓	✓	×	97.25	94.67	90.83	83.21	81.48	68.75	94.07	88.81
✓	×	✓	97.36	94.86	91.06	83.60	82.18	69.76	94.26	89.14
×	✓	✓	97.30	94.75	90.98	83.47	81.52	68.82	94.14	88.91
✓	✓	✓	**97.97**	**96.03**	**91.64**	**84.56**	**82.42**	**70.11**	**94.82**	**90.16**

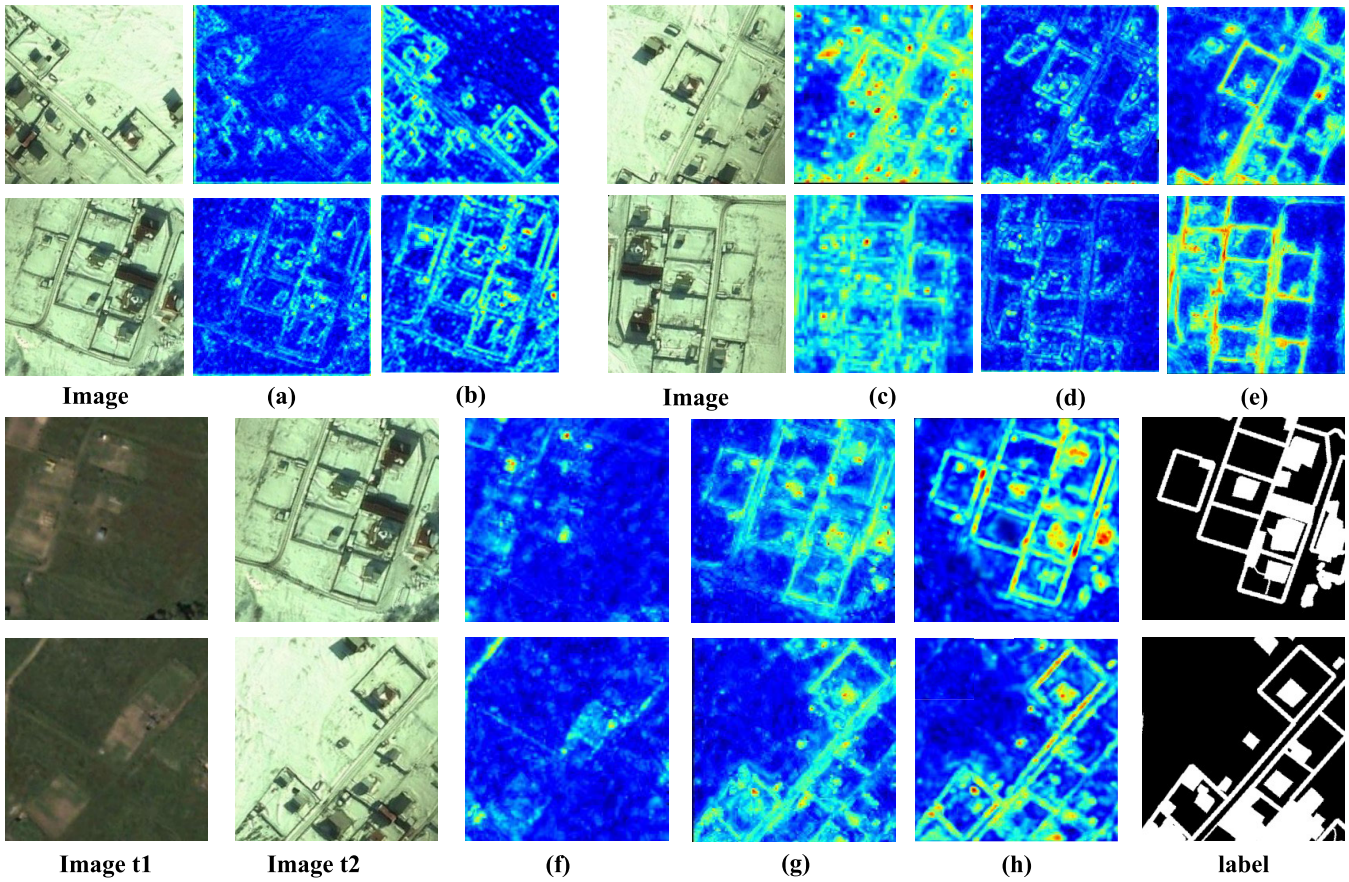


Fig. 10. Feature visualization. (a) Input of the MACA module. (b) Output of the MACA module. (c) and (d) Two inputs of the DDA module. (e) Output of the DDA module. (f) and (g) Two inputs of the DG module. (h) Output of the DG module. Red indicates higher attention values and blue indicates lower attention values.

the effectiveness of submodules in the DG block. Through fusing spatial, channel, and difference attention information, the DG module can provide more comprehensive guidance for CD tasks, enabling the model to fully integrate and process variations.

F. Feature Visualization

The MACA module is used to achieve multiscale information attention effectively, and Fig. 10(a) and (b) shows the visualization to further demonstrate its effectiveness. Specifically, Fig. 10(a) represents the feature map before input to the MACA module, and Fig. 10(b) represents the output feature map after the MACA module. As we envisioned, the MACA module can effectively extract strip buildings and refine building contours, which provides rich semantic information for image upsampling. In the upsampling stage, we introduce the DDA module to achieve efficient fusion of features, and Fig. 10(c)–(e) shows the visualization results. As displayed, Fig. 10(c) represents the deep features, Fig. 10(d) represents

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TABLE VI
ABLATION STUDY ON DG MODULE

DG Module			CDD		LEVIR-CD		SYSU-CD		BCDD	
PAM	CAM	DIM	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
×	×	×	97.28	94.69	90.79	83.14	81.53	68.82	94.11	88.86
✓	×	×	97.47	95.09	91.05	83.57	81.82	69.23	94.28	89.18
✓	✓	×	97.69	95.48	91.31	84.03	82.08	69.61	94.46	89.51
✓	✓	✓	**97.97**	**96.03**	**91.64**	**84.56**	**82.42**	**70.11**	**94.82**	**90.16**

TABLE VII
ABLATION STUDY ON DIFFERENT ATTENTION MODULES

ATTENTION	Module	Params.(B)	CDD		LEVIR-CD		SYSU-CD		BCDD	
			F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
Feature only	SAM	90	97.70	95.51	91.25	83.91	81.73	69.11	93.98	88.65
Feature only	CAM	32	97.61	95.35	91.11	83.68	81.90	69.36	94.67	89.90
Feature only	CBAM	50	97.74	95.58	91.18	83.80	81.34	68.55	93.69	88.13
Convolution only	Dyconv	48	97.65	95.43	91.33	84.04	81.52	68.81	94.43	89.46
Convolution&Feature	(our DDA)	64	**97.97**	**96.03**	**91.64**	**84.56**	**82.42**	**70.11**	**94.82**	**90.16**

the shallow features, and Fig. 10(e) represents the feature map after fusion by the DDA module. It shows that deep features and shallow features contain different information of the image, which leads to redundant features. The DDA module can perform adaptive fusion in both feature and convolution dimensions, and the fused image focuses more on foreground features such as buildings and roads. To drive the network spotlight on the real changed regions, we propose the DG module to reduce the effect of pseudo-change caused by illumination and noise, Fig. 10(f)–(h) shows this process. Fig. 10(f) represents the image features at moment T_1, Fig. 10(g) represents the image features at moment T_2, and Fig. 10(h) represents the feature maps obtained after the DG module. The parallel features from the bi-temporal images are guided through DG module, and the obtained features are closer to the real variations. Therefore, the final combined model can get better detection results.

V. DISCUSSION

In this section, we evaluate the effectiveness of DDA module and MANet, respectively, then visualize and analyze the training process. Finally, effects of different loss functions on MANet are discussed.

A. Analysis of DDA Module

Feature-based attention is performed on the feature map, which can be simply divided into spatial and channel attentions. Convolution-based attention is the allocation of attention to convolutional kernels. Feature attention and convolutional attention are two important branches of attention mechanisms. However, existing studies mainly divide them into two separate parts. To explore the deeper relationship between them, we propose the attention module that combines both feature and convolution, i.e., the DDA module. Fig. 11 shows the difference between our module and existing attentions.

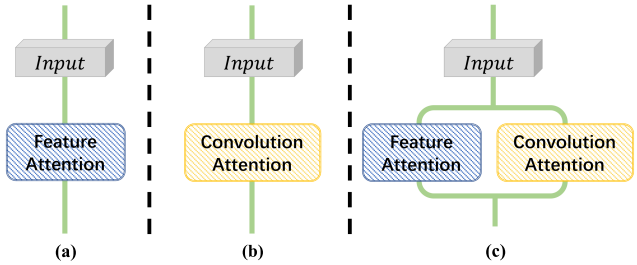


Fig. 11. DDA module. (a) Feature based. (b) Convolution based. (c) Our DDA.

We experiment with different attention mechanisms instead of the DDA module on different datasets under the same conditions, respectively. These include spatial attention module (SAM), CAM, CBAM, and dynamic convolution (Dyconv) [52], [56], [60], [61]. As shown in Table VII, we visualize the parameters for each attention module. SAM has a maximum parameter count of 90 bytes, which results from its attention on the computation of spatial dimensions. CAM has a minimum parameter number for 32 bytes, which due to its focus only on the relationships between channels. This is also the reason that CAM ignores the contribution of information in the spatial dimension, which results in the lowest IoU in CDD and

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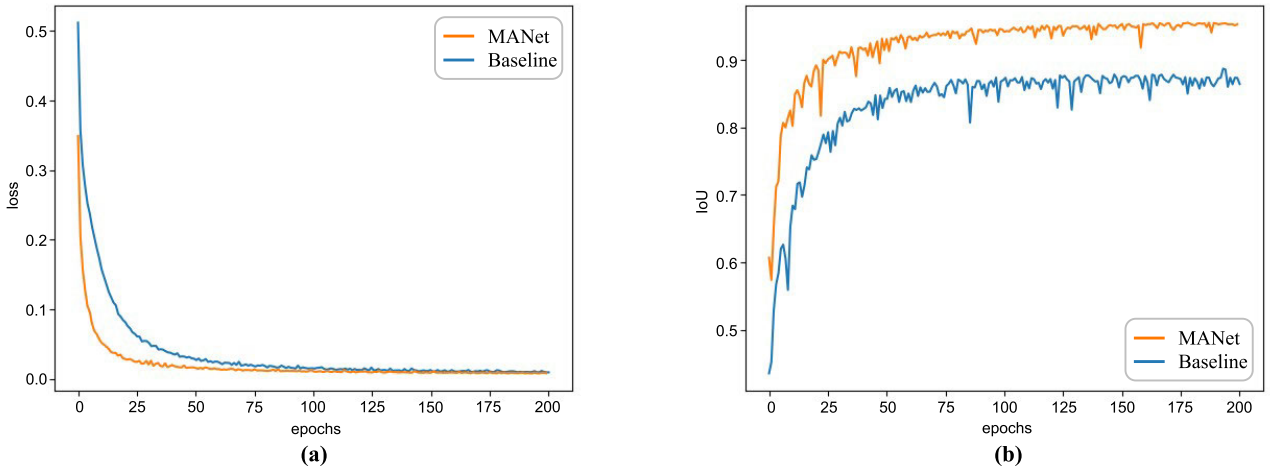


Fig. 12. Metrics variation of models for each training epoch on CDD dataset. (a) Training loss. (b) Validation IoU.

LEVIR-CD datasets. CBAM uses lightweight hybrid attention mechanism that can focus on both spatial and channel features, but it only improves IoU by 0.07% and 0.23% in CDD dataset compared to SAM and CAM. In the convolutional dimension, Dyconv parameter number is 48 bytes, which improves the IoU by 0.08% compared to CAM in CDD dataset. The proposed DDA has 64 bytes of parameters, which is slightly higher than CBAM and Dyconv, but significantly less than SAM. Considering the network parameter scales as MB, the designed DDA module has a very limited impact on the network compared to the above attention mechanism. In addition, DDA achieves optimal performance in all four datasets, which is a substantial improvement compared to others. The IoU of DDA is improved from 0.45% to 0.68% over other modules on the CDD dataset, 0.52% to 0.88% on the LEVIR-CD dataset, 0.75% to 1.56% on the SYSU-CD dataset, and 0.26% to 2.03% on the BCDD dataset. This demonstrates that DDA can better capture the semantic representation in both feature and convolutional dimensions, thus achieving the best detection performance.

B. Metrics Visualization

Fig. 12 illustrates the loss and IoU variation on the training/validation set for each training epoch. From Fig. 12(a), it is evident that MANet has a lower initial loss and can achieve the minimum loss faster compared to the baseline model, which indicates MANet can capture the data features more efficiently. Fig. 12(b) presents the IoU variation curve, which shows the IoU of MANet increases faster than the baseline model and has a higher upper threshold. This demonstrates that MANet can achieve better performance on the validation dataset while having better convergence and generalization to the dataset.

C. Network Efficiency

Table VIII shows the results of the comparison of model efficiency, using model parameter counts (Params) and floating-point operations per second (FLOPs) to evaluate network complexity and computational efficiency. Three variants based on UNet have a smaller parameter size. However, they have the lowest detection accuracy. This is due to their simpler model structure, which helps to reduce computational resource usage but also limits the feature representation capability of networks. In comparison, our model is higher than SRCDNet, BITNet, slightly higher than ISNet, DSAMNet and UGSNet, lower than DDCNN and DETNet in parameter number, basically maintaining the same parameter level. For FLOPs, our computational cost is lower than DSAMNet and DETNet, much lower than DDCNN. Overall, MANet is at a moderate level in parameter count and computation speed, but it achieves more superior outcomes, improving IoU by 5.38% compared to DDCNN in CDD dataset. Therefore, MANet achieves a significant improvement in detection performance with a relatively appreciable parameter and computational speed, which is associated with the proposed design strategy. We still need to further explore the potential of lightweight network architectures in CD field.

TABLE VIII
SPACE COMPLEXITY VERSUS PERFORMANCE ON CDD DATASET

Methods	Param.(M)	FLOPs.(G)	F1(%)	IoU(%)
FC-EF	1.35	3.56	73.11	57.62
FC-Siam-diff	1.82	4.71	82.94	70.85
FC-Siam-conc	2.03	5.32	84.12	72.60
SRCDNet	24.37	12.03	92.96	86.85
BITNet	22.87	26.31	94.15	88.95
ISNet	36.64	8.49	95.16	90.78
DSAMNet	33.85	75.39	93.81	88.35
UGSNet	32.09	26.67	94.39	89.38
DDCNN	46.68	135.78	94.33	90.65
DETNet	58.62	60.46	95.12	90.71
Ours	38.61	52.42	97.97	96.03

D. Experiments on the Loss Function

To validate the efficiency of loss function, we introduce WCE loss function for ablation experiments [41]. As shown

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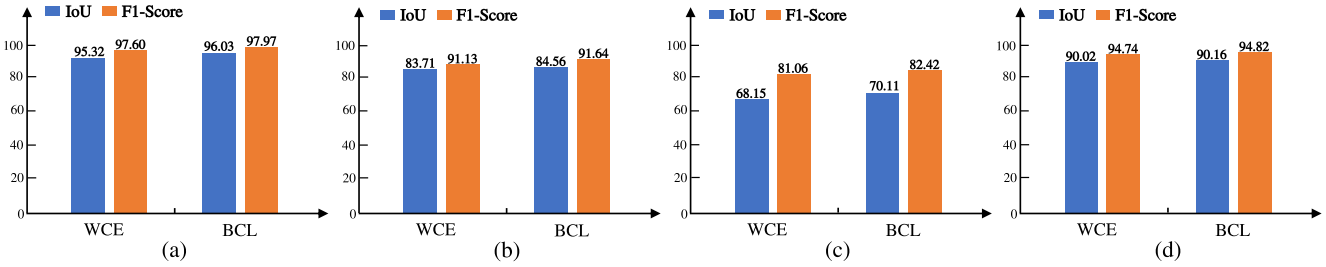


Fig. 13. Ablation experiments of loss function. (a) CDD dataset. (b) LEVIR-CD dataset. (c) SYSU-CD dataset. (d) BCDD dataset.

in Fig. 13, BCL achieves better performance on the CDD, LEVIR-CD, SYSU-CD, and BCDD datasets. Compared to WCE, MANet with BCL improves IoU and $F1$ by 0.71% and 0.37% on the CDD dataset, 0.85% and 0.51% on the LEVIR-CD dataset, 1.96% and 1.36% on the SYSU-CD dataset, 0.14% and 0.08% on the BCDD dataset. Overall, BCL can adaptively adjust sample weights to better handle the category imbalance problem. In addition, BCL introduces boundary values to limit the distance range of positive and negative samples, which enhances the generalization ability of the model. Therefore, BCL has better adaptability than WCE in handling category imbalance problems, which improves model performance and achieves accurate CD results.

VI. Conclusion

In this article, we present a novel CD network, namely MANet. The proposed network alleviates the major problem of the popular Siamese CNN through the introduction of three modules: MACA, DDA, and DG module. MACA module can effectively realize the extraction of multiscale information and refine the object contour. The DDA module enables efficient fusion of deep and shallow features, thus enhancing feature representation and reducing feature redundancy. The DG module reduces the influence of irrelevant variations on the prediction outcomes and guides the network to learn the real changed regions. Experimental results demonstrate our presented method has an excellent robustness and generality, which achieves optimal results on four popular CD datasets. However, MANet still has some limitations. First, it is a binary detection method and cannot be used for multiclass CD tasks. Second, when dealing with complex multisource RS images, such datasets with large spatial differences remain a significant challenge for MANet. Next, we shall discover more efficient CD algorithms to address the challenges of diverse scenes.

References

[1] N. Goyette, P.-M. Jodoin, F. Porikli, J. Konrad, and P. Ishwar, "Changedetection. Net: A new change detection benchmark dataset," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2012, pp. 1–8.

[2] S. Jin, L. Yang, P. Danielson, C. Homer, J. Fry, and G. Xian, "A comprehensive change detection method for updating the national land cover database to circa 2011," *Remote Sens. Environ.*, vol. 132, pp. 159–175, May 2013.

[3] D. Brunner, G. Lemoine, and L. Bruzzone, "Earthquake damage assessment of buildings using VHR optical and SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 5, pp. 2403–2420, May 2010.

[4] H. Hou, R. C. Estoque, and Y. Murayama, "Spatiotemporal analysis of urban growth in three African capital cities: A grid-cell-based analysis using remote sensing data," *J. Afr. Earth Sci.*, vol. 123, pp. 381–391, Nov. 2016.

[5] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 91–106, Jun. 2013.

[6] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.

[7] H. Mahmoudzadeh, "Digital change detection using remotely sensed data for monitoring green space destruction in Tabriz," *Int. J. Environ. Res.*, vol. 1, no. 1, p. 411735, 2007.

[8] A. Shafique, G. Cao, Z. Khan, M. Asad, and M. Aslam, "Deep learning-based change detection in remote sensing images: A review," *Remote Sens.*, vol. 14, no. 4, p. 871, Feb. 2022.

[9] T. Habib, J. Inglada, G. Mercier, and J. Chanussot, "Support vector reduction in SVM algorithm for abrupt change detection in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 606–610, Jul. 2009.

[10] G. Verdier and A. Ferreira, "Adaptive Mahalanobis distance and k-nearest neighbor rule for fault detection in semiconductor manufacturing," *IEEE Trans. Semicond. Manuf.*, vol. 24, no. 1, pp. 59–68, Feb. 2011.

[11] S. E. Sesnie, P. E. Gessler, B. Finegan, and S. Thessler, "Integrating Landsat TM and SRTM-DEM derived variables with decision trees for habitat classification and change detection in complex neotropical environments," *Remote Sens. Environ.*, vol. 112, no. 5, pp. 2145–2159, May 2008.

[12] O. A. C. Júnior et al., "A new approach to change vector analysis using distance and similarity measures," *Remote Sens.*, vol. 3, no. 11, pp. 2473–2493, Nov. 2011.

[13] Q. Liu, G. Liu, C. Huang, S. Liu, and J. Zhao, "A tasseled cap transformation for Landsat 8 OLI TOA reflectance images," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 541–544.

[14] A. Ghosh, N. S. Mishra, and S. Ghosh, "Fuzzy clustering algorithms for unsupervised change detection in remote sensing images," *Inf. Sci.*, vol. 181, no. 4, pp. 699–715, Feb. 2011.

[15] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. remote Sens.*, vol. 60, 2021, Art. no. 5604816.

[16] E. A. Addink, F. M. B. Van Coillie, and S. M. De Jong, "Introduction to the GEOBIA 2010 special issue: From pixels to geographic objects in remote sensing image analysis," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 15, pp. 1–6, Apr. 2012.

[17] J. Cai, J. Li, W. Li, and J. Wang, "Deeplearning model used in text classification," in *Proc. 15th Int. Comput. Conf. Wavelet Act. Media Technol. Inf. Process. (ICCWAMTIP)*, 2018, pp. 123–126.

[18] Y. Cui, L. Yan, Z. Cao, and D. Liu, "TF-blender: Temporal feature blender for video object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8138–8147.

[19] Z. Jin et al., "Mining contextual information beyond image for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jul. 2021, pp. 7231–7241.

[20] D. Hong, J. Yao, D. Meng, Z. Xu, and J. Chanussot, "Multimodal GANs: Toward crossmodal hyperspectral–multispectral image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5103–5113, Jun. 2021.

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[21] J. Cao, Z. Chen, and B. Wang, "Graph-based deep convolutional networks for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3270–3273.

[22] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Auto. Robots*, vol. 42, no. 7, pp. 1301–1322, Oct. 2018.

[23] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[24] M. Villa, G. Dardenne, M. Nasan, H. Letissier, C. Hamitouche, and E. Stindel, "FCN-based approach for the automatic segmentation of bone surfaces in ultrasound images," *Int. J. Comput. Assist. Radiol. Surgery*, vol. 13, no. 11, pp. 1707–1716, Nov. 2018.

[25] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[26] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4403718.

[27] M. Wang, K. Tan, X. Jia, X. Wang, and Y. Chen, "A deep Siamese network with hybrid convolutional feature extraction module for change detection based on multi-sensor remote sensing images," *Remote Sens.*, vol. 12, no. 2, p. 205, Jan. 2020.

[28] K. Jiang, J. Liu, F. Liu, W. Zhang, Y. Liu, and J. Shi, "Dual UNet: A novel Siamese network for change detection with cascade differential fusion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 1428–1431.

[29] X. Hou, Y. Bai, Y. Li, C. Shang, and Q. Shen, "High-resolution triplet network with dynamic multiscale feature for change detection on satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 103–115, Jul. 2021.

[30] P. Chen, B. Zhang, D. Hong, Z. Chen, X. Yang, and B. Li, "FCCDN: Feature constraint network for VHR image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 101–119, May 2022.

[31] T. Lei et al., "Difference enhancement and spatial–spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4507013.

[32] M. Gong, Y. Liang, J. Shi, W. Ma, and J. Ma, "Fuzzy C-means clustering with local information and kernel metric for image segmentation," *IEEE Trans. Image Process.*, vol. 22, no. 2, pp. 573–584, Feb. 2013.

[33] L. Bruzzone and D. F. Prieto, "A minimum-cost thresholding technique for unsupervised change detection," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3539–3544, Jan. 2000.

[34] R. Suganya and R. Shanthi, "Fuzzy C-means algorithm—A review," *Int. J. Sci. Res. Publications*, vol. 2, no. 11, pp. 1–3, 2012.

[35] S. P. Chatzis and T. A. Varvarigou, "A fuzzy clustering approach toward hidden Markov random field models for enhanced spatially constrained image segmentation," *IEEE Trans. Fuzzy Syst.*, vol. 16, no. 5, pp. 1351–1361, Oct. 2008.

[36] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2015, pp. 3431–3440.

[37] Z. Lv, H. Huang, L. Gao, J. A. Benediktsson, M. Zhao, and C. Shi, "Simple multiscale UNet for change detection with heterogeneous remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 2504905.

[38] Z. Zhou, M. M. Rahman Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-net architecture for medical image segmentation," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, 2018, pp. 3–11.

[39] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021.

[40] H. Chen, Y. Zao, L. Liu, S. Chen, and Z. Shi, "Semantic decoupled representation learning for remote sensing image change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1051–1054.

[41] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 8007805.

[42] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5920416.

[43] G. Cheng, G. Wang, and J. Han, "ISNet: Towards improving separability for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623811.

[44] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2021.

[45] Q. Li, M. Gong, Y. Yuan, and Q. Wang, "Symmetrical feature propagation network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536912.

[46] Q. Li, Y. Yuan, X. Jia, and Q. Wang, "Dual-stage approach toward hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 31, pp. 7252–7263, 2022.

[47] Q. Li, M. Gong, Y. Yuan, and Q. Wang, "RGB-induced feature modulation network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512611.

[48] K. Jiang, W. Zhang, J. Liu, F. Liu, and L. Xiao, "Joint variation learning of fusion and difference features for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4709918.

[49] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, 2016, pp. 770–778.

[50] Y. Qian and P. C. Woodland, "Very deep convolutional neural networks for robust speech recognition," in *Proc. IEEE Spoken Lang. Technol. Workshop (SLT)*, Dec. 2016, pp. 481–488.

[51] X. Ding, Y. Guo, G. Ding, and J. Han, "ACNet: Strengthening the kernel skeletons for powerful CNN via asymmetric convolution blocks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jul. 2019, pp. 1911–1920.

[52] Y. Chen, X. Dai, M. Liu, D. Chen, L. Yuan, and Z. Liu, "Dynamic convolution: Attention over convolution kernels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11027–11036.

[53] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on imagenet classification," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 1026–1034.

[54] R. Hadsell, S. Chopra, and Y. LeCun, "Dimensionality reduction by learning an invariant mapping," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 2, 2006, pp. 1735–1742.

[55] M. Lebedev, Y. V. Vizilter, O. Vygolov, V. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, no. 2, pp. 1–7, 2018.

[56] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[57] K. Wu, Q. Du, Y. Wang, and Y. Yang, "Supervised sub-pixel mapping for change detection from remotely sensed images with different resolutions," *Remote Sens.*, vol. 9, no. 3, p. 284, Mar. 2017.

[58] D. He, Q. Shi, X. Liu, Y. Zhong, G. Xia, and L. Zhang, "Generating annual high resolution land cover products for 28 metropolises in China based on a deep super-resolution mapping network using Landsat imagery," *GIScience Remote Sens.*, vol. 59, no. 1, pp. 2036–2067, Dec. 2022.

[59] Q. Shi, M. Liu, A. Marinoni, and X. Liu, "UGS-1m: Fine-grained urban green space mapping of 31 major cities in China based on the deep learning framework," *Earth Syst. Sci. Data*, vol. 15, no. 2, pp. 555–577, Feb. 2023.

[60] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[61] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "Cbam: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.



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